A beginner's guide to Collagen Fibre detection & Collagen Fibre Orientation Disorder (CFOD)

Gaussians are separable

Derivative of Gaussians: DtG^{pq}

filter banks: DtG^{00} → zeroth order

$$DtG \begin{cases} DtG^{10}, DtG^{01} & \rightarrow \text{First order} \\ DtG^{20}, DtG^{11}, DtG^{02} & \rightarrow 2^{nd} \text{ order} \end{cases}$$

i in [(0,0), (1,0) . . . . . (0,2)]

jet[i,] = Convolution(img, DtG[i,0,:], DtG[i,1,:]) $\times \sigma^{(p+q)}$

$$\lambda\_val = jet[3] + jet[5]$$

$$\mu = \sqrt{(jet[3] - jet[5])^2 + (4 \times jet[4]^2)}$$

c[:, :, 0] = epsilon × jet[0]

$2 \times \sqrt{jet[1]^2 + jet[2]^2}$

λ_val

$-\lambda\_val$

$\frac{1}{\sqrt{2}}(\mu + \lambda\_val)$

$\frac{1}{\sqrt{2}}(\mu - \lambda\_val)$

why? (STUDY MORE MATHS)

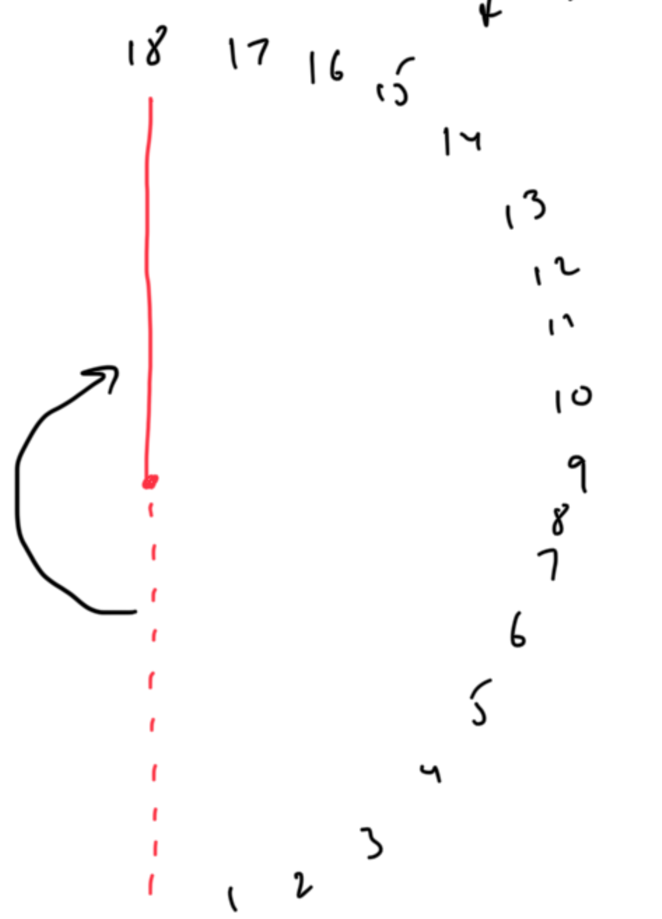
c[:, :, 6]   μ

BIF = argmax(c, axis=2)

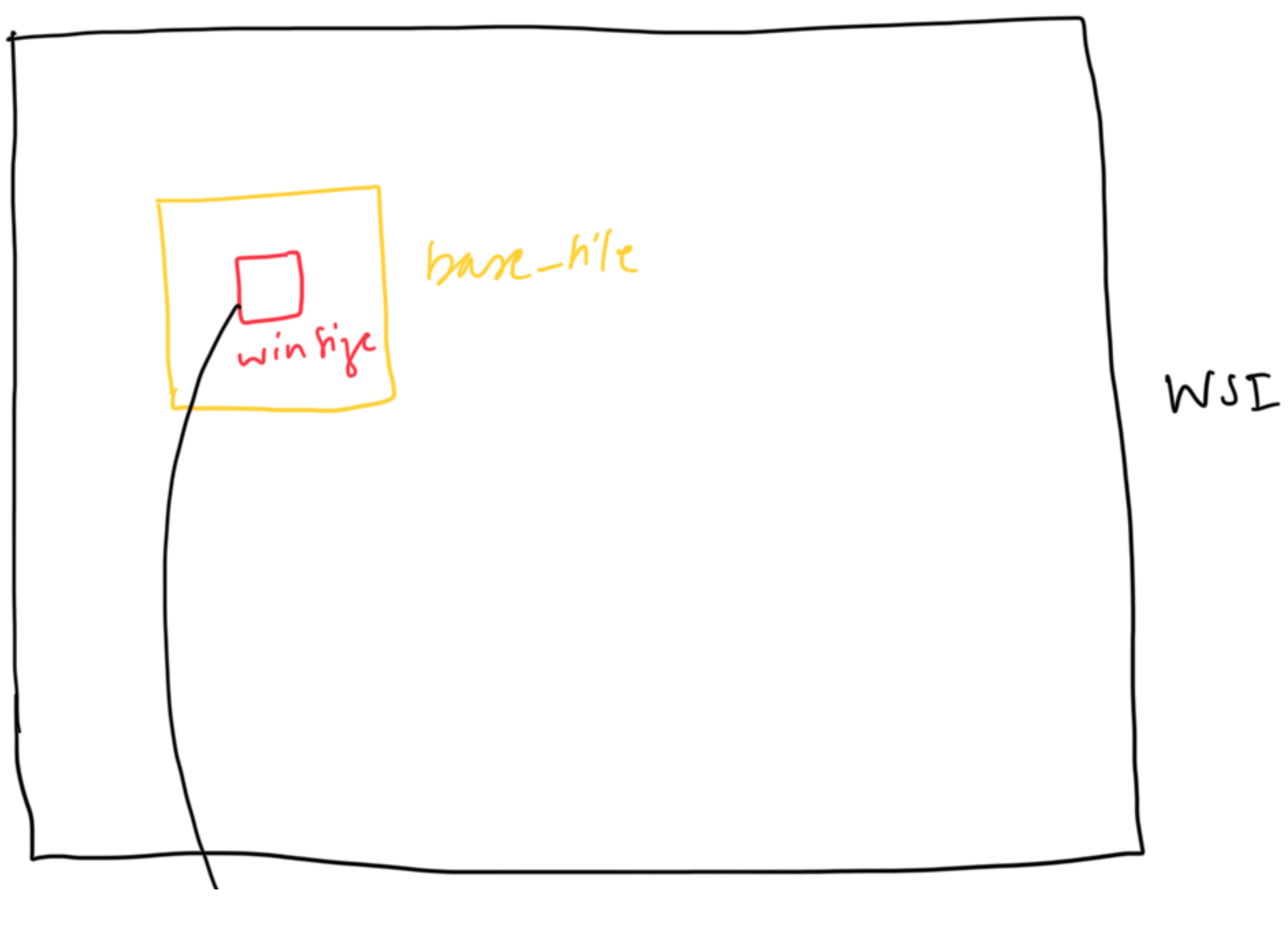
Collagen fibre is the 5th channel

collagen fibre
- centroid
- Area
- Orientation

bin



0th & 18th are same so convert 0th → 18th



If it has collagen fibres more than a threshold
(use centroid to check)

Now for disorderness there are two types of pairs

[3, 3, 3, 1, 1, 2] orientation

[a, b, c, d, d, f] area

p[ ] → co-occurrence matrix

different orientations

3 — 1

3 — 2

1 — 2

p[3,1] = (a+b+c) × (d+d)

p[3,2] = (a+b+c) × f

p[1,2] = (d+d) × f

same orientations

3 3 3
a b c

a — b

a — c

b — c

p[3,3] = ab + ac + bc

(can't do (a+b+c) × (a+b+c))

1 1
d d

d — d

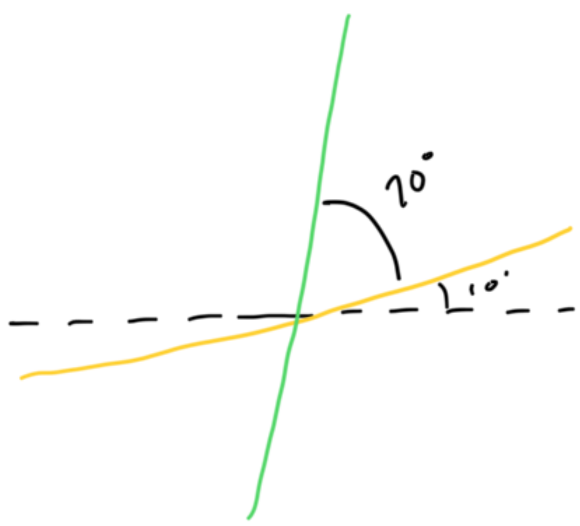
p[1,1] = dd

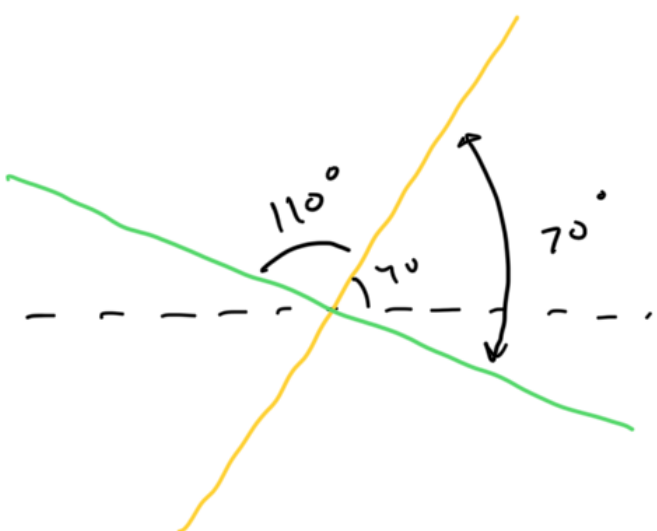
angle b/w a pair of CF

$\text{all\_contrast} \rightarrow$ angle b/w a pair of CF

Collagen fibre with x & $18-x$ between them are same
(a° & $180 - a^\circ$)

$\underbrace{1 \quad 8}_{7}$ $\qquad$ 4 $\quad$ 15 ''

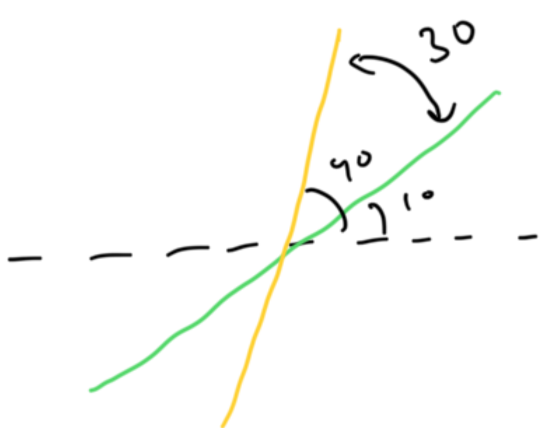


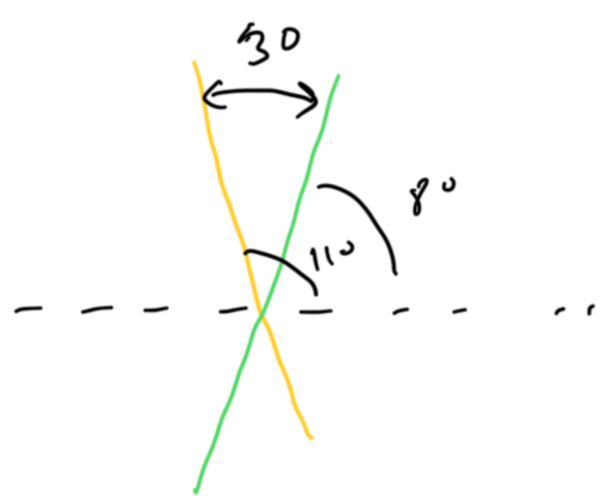


$\text{all-contrast} \rightarrow$ 1, 8, 10, 2, 5, 14, 17 $\rightarrow$ 1, 8, 8, 2, 5, 4, 1

weighted co-occurrence P_1 P_2 P_3 P_4 P_5 P_6 P_7 $\qquad$ P_1 P_2 P_3 P_4 P_5 P_6 P_7

∴ add the weighted co-occurrence of collagen fibre pairs with same angle between them





$$1, \quad 8, \quad 2, \quad 5, \quad 4$$

$$p_1 + p_7 \quad p_2 + p_3 \quad p_4 \quad p_5 \quad p_6$$

Contrast entropy $\rightarrow -\sum p \log p$